Illustration:

An electric power of 2 MW is transmitted to a place through transmission lines of total resistance R = 40 Ω, at two different voltages. One is lower voltage (10 kV) and the other is higher (100 kV). Let us now calculate and compare power losses in these two cases.

Case (i):

P = 2 MW; R = 40 Ω; V = 10 kV

Power, $P = VI$

$$\therefore \text{Current, } I = \frac{P}{V}$$

$$= \frac{2\times10^6}{10\times10^3} = 200\text{A}$$

Power loss = Heat produced = I^2R

$$= (200)^2 \times 40 = 1.6\times10^6\,\text{W}$$

$$\% \text{ of power loss} = \frac{1.6\times10^6}{2\times10^6}\times100\%$$

$$= 0.8\times100\% = 80\%$$

Case (ii):

P = 2 MW; R = 40 Ω; V = 100 kV

$$\therefore \text{ Current, } I = \frac{P}{V}$$

$$= \frac{2\times10^6}{100\times10^3} = 20\text{A}$$

Power loss = I^2R

$$= (20)^2 \times 40 = 0.016\times10^6\,\text{W}$$

$$\% \text{ of power loss} = \frac{0.016\times10^6}{2\times10^6}\times100\%$$

$$= 0.008\times100\% = 0.8\%$$

Thus it is clear that when an electric power is transmitted at higher voltage, the power loss is reduced to a large extent.

EXAMPLE 4.16

An ideal transformer has 460 and 40,000 turns in the primary and secondary coils respectively. Find the voltage developed per turn of the secondary if the transformer is connected to a 230 V AC mains. The secondary is given to a load of resistance 10^4 Ω. Calculate the power delivered to the load.

Solution

N_P = 460 turns; N_S = 40,000 turns

V_P = 230 V; $R_S = 10^4$ Ω

(i) Secondary voltage,

$$V_S = \frac{V_P N_S}{N_P} = \frac{230\times40{,}000}{460} = 20{,}000\,\text{V}$$

$$\text{Secondary voltage per turn, } \frac{V_S}{N_S} = \frac{20{,}000}{40{,}000}$$

$$= 0.5\ \text{V}$$

(ii) Power delivered

$$= V_S I_S = \frac{V_S^2}{R_S} = \frac{20{,}000\times20{,}000}{10^4} = 40\ \text{kW}$$

EXAMPLE 4.17

An inverter is common electrical device which we use in our homes. When there is no power in our house, inverter gives AC power to run a few electronic appliances like fan or light. An inverter has inbuilt step-up transformer which converts 12 V AC to 240 V AC. The primary coil has 100 turns and the inverter delivers 50 mA to the external circuit. Find the number of turns in the secondary and the primary current.

Solution

V_p = 12 V; V_s = 240 V

I_s = 50mA; N_p = 100 turns

$$\frac{V_S}{V_P} = \frac{N_S}{N_P} = \frac{I_P}{I_S} = K$$

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